The documentation and process conversion measures necessary to comply with this revision shall be completed by 20 June 1995

INCH-POUND
MIL-S-19500/302B
20 January 1995
SUPERSEDING
MIL-S-19500/302A
20 August 1969

MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN,
SILICON, LOW-POWER,
TYPES 2N2708, JAN

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, VHF-UHF amplifier transistor. One level of product assurance is provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figures 1 (similar to a TO-72).

1.3 Maximum ratings.

P_T 1/ T_A = +25°C	P_T 2/ T_C = +25°C	I_C	V_{CBO}	V_{CEO}	V_{EBO}	T_J and T_{STG}
mW	mW	mA dc	V dc	V dc	V dc	°C
300	200	50	35	20	3	-65 to +200

1/ Derate linearly 1.71 mW/°C for TC > +25°C.
2/ Derate linearly 1.14 mW/°C for TA > +25°C

1.4 Primary electrical characteristics at T_A = +25°C.

	h_{FE} I_C = 2 mA dc V_{CE} = 2 V dc	$\|h_{fe}\|$ I_C = 1.0 mA dc V_{CE} = 15 V dc f = 100 MHz	$r_b'C_c$ I_C = 2 mA dc V_{CB} = 2 V dc f = 31.8 MHz	C_{cb} V_{CB} = 15 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	NF V_{CE} = 15 V dc I_C = 2 mA dc f = 200 MHz	G_{pe} V_{CE} = 15 V dc I_C = 2 mA dc f = 200 MHz
			psec	pf	db	db
Min	30	7	9			15
Max	180	12	33	1.0	7.5	22

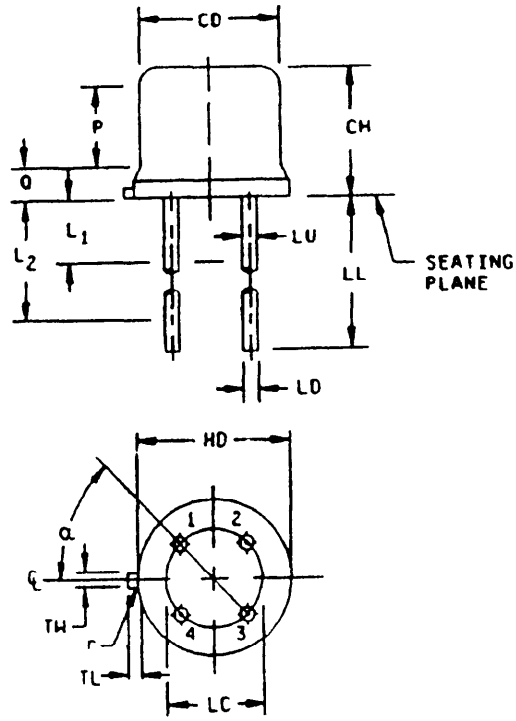
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Research Laboratory Electronics and Power Sources Directorate, ATTN: AMSRL-EP-RD, Ft. Monmouth, NJ 07703-5601 by using the Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5961

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Symbol	Inches		Millimeters		Note
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54		6
LD	.016	.021	0.41	0.53	7
LL	.500	.750	12.70	19.05	7
LU	.016	.019	0.41	0.48	7
L_1	---	.050	---	1.27	7
L_2	.250	---	6.35	---	7
TL	.028	.048	0.71	1.22	3
TW	.036	.046	0.91	1.17	10
P	.100	---	2.54	---	
Q	---	.040	---	1.02	4
r	---	.007	---	0.18	11
α	45° TP				6



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Symbol TL is measured from HD maximum.
4. Details of outline in this zone are optional.
5. Symbol CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 inch (1.37 mm) +.001 inch (0.03 mm) -.000 inch (0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of TP relative to tab. Device may be measured by direct methods or by gauge.
7. Symbol LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Lead diameter shall not exceed .042 inch (1.07 mm) within L_1 and beyond LL minimum.
8. Lead designation, depending on device type, shall be as follows: 1 - emitter, 2 - base, 3 - collector, 4 - Case.
9. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions (similar to TO-72).

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document take precedence. Nothing in this document, however, supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in figure 1 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in Mil-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.3 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.3.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

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4.3.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in IVb (JAN) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
B3	1026	V_{CB} = 15 V dc; P_T = 200 mW at T_A = +25° ±3°C;
B5	----	Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Test condition E.
C6	1026	V_{CB} = 15 V dc; P_T = 200 mW at T_A = +25° ±3°C;

4.4 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.4.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.4.2 Collector-base feedback capacitance. This test shall be conducted in accordance with method 3236 of MIL-STD-750, except that the case and emitter leads shall be connected to the guard terminal of the measuring equipment.

4.4.3 Noise figure test. Noise figure shall be measured using a HP 342A voltmeter and 343A noise source in conjunction with the small signal power gain test circuit or equivalent.

4.4.4 Disposition of "CASE" lead. Except for the following tests, the "CASE" lead shall be "floating" during all electrical tests and measurements: noise figure; small-signal power gain; collector-base time constant; and magnitude of $|h_{fe}|$. These tests shall be conducted with the "CASE" lead connected to a common rf chassis ground.

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TABLE I. Group A inspection

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 1.0 uA dc	$V_{(BR)CBO}$	35		V dc
Breakdown voltage, emitter to base	3026	Bias condition D; I_E = 10 uA dc	$V_{(BR)EBO}$	3		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.4.1)	$V_{(BR)CEO}$	20		V dc
Collector to base cutoff Current	3036	Bias condition D; V_{CB} = 15 V dc	I_{CBO}		10	nA dc
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 2 mA dc	h_{FE}	30	180	
Collector-emitter saturation voltage	3071	I_C = 10 mA dc; I_B = 1 mA dc	$V_{CE(sat)}$		0.4	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 10 mA dc; I_B = 1 mAdc;	$V_{BE(sat)}$		1.0	V dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 15 V dc	I_{CBO}		1.0	µA dc
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 2.0 mA dc	h_{FE}	15		
Subgroup 4						
Small-signal short-circuit forward current transfer ratio	3206	V_{CE} = 15 V dc; I_C = 2 mA dc;	h_{fe}	30	180	
Magnitude of small-signal short-circuit forward current transfer ratio	3306	V_{CE} = 15 V dc; I_C = 2 mA dc; f = 100 MHz	$\lvert h_{fe} \rvert$	7	12	

See footnotes at end of table.

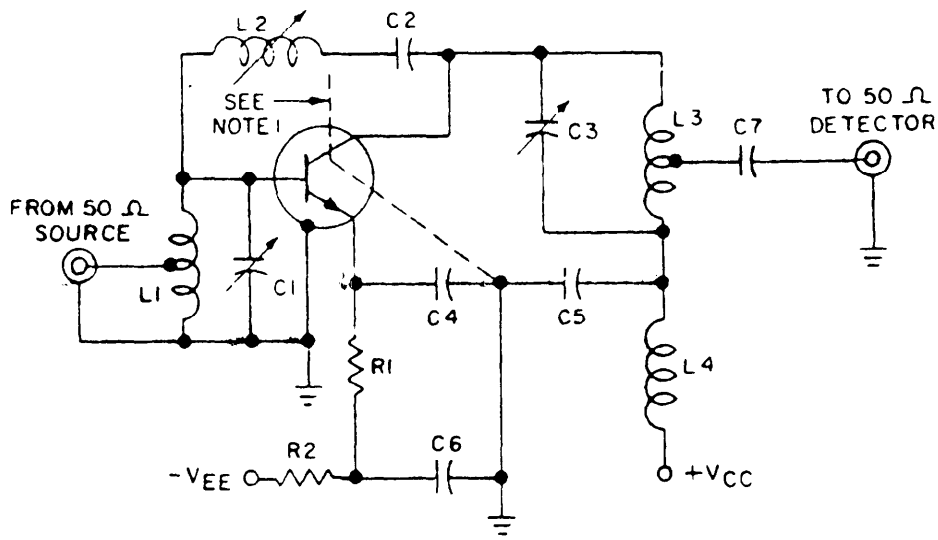
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TABLE I. Group A inspection - continued.

Inspection 1/ 2/	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limit Min	Limit Max	Unit
Subgroup 4 - continued						
Collector-base feedback capacitance (see 4.4.2)	3236	V_{CB} = 15 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{cb}		1	pF
Subgroup 5						
Small-signal power gain circuited)	----	V_{CE} = 15 V dc; I_C = 2 mA dc f = 200 MHz (see 4.4.4 and figure 2)	G_{pe}	15	22	db
Collector-base time constant	----	V_{CB} = 15 V dc I_C = 2 mA dc f = 31.8 MHz (see 4.4.4 and figure 3)	$r_b'C_c$	9	33	ps
Noise figure	3246	V_{CE} = 15 V dc I_C = 2 mA dc f = 200 MHz R_g = 50 Ω (see 4.4.3 and 4.4.4)	NF		7.5	db
Subgroups 6 and 7						
Not required						

1/ For sampling plan see MIL-S-19500.

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NOTE 1: External interlead shield to isolate the collector from the emitter and base leads.

NEUTRALIZATION PROCEDURE:

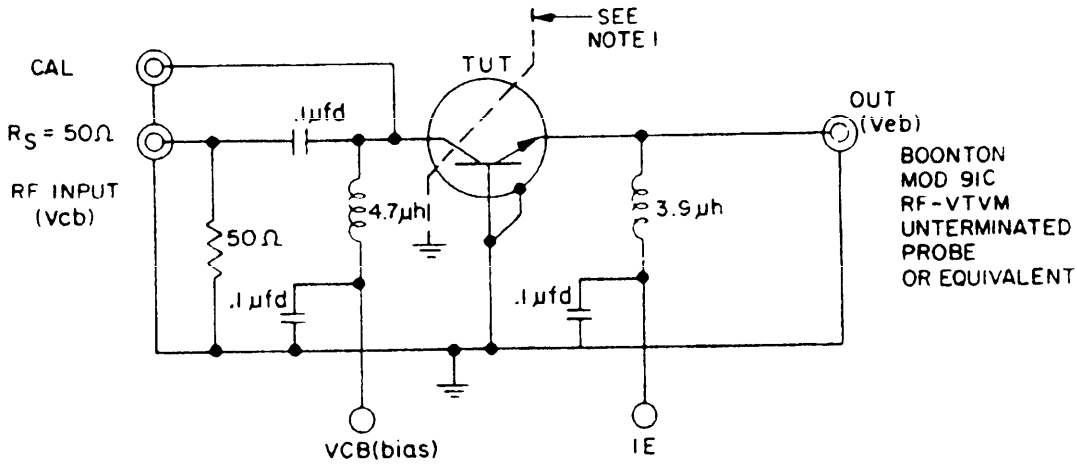
1. Connect a 200 MHz signal generator (with a 50 ohm output impedance) to the input terminals of the amplifier, and connect a 50 ohm rf voltmeter to the output terminals of the amplifier.
2. Apply V_{EE} and V_{CC} to obtain the specified conditions.
3. Adjust the output of the signal generator to approximately 10 millivolts and tune C_1 and C_3 for maximum output.
4. Interchange the connections to the signal generator and rf voltmeter and with sufficient signal applied at the output terminals, tune L_2 for a minimum indication on the rf voltmeter.
5. Repeat this sequence until optimum settings are obtained for all variables.

Circuit-component information:

C_1	3 - 12 pF
C_2 and C_7	1000 pF
C_3	1.5 - 7.5 pF
C_4 and C_5	0.01 µF
C_6	0.05 µF
L_1	3.5 T No. 16 AWG 5/16 ID, 7/16 length, Turns ratio = 2 to 1
L_2	0.4 - 0.65 µh, Miller No. 4303 or equal.
L_3	8 T No. 16 AWG, 1/8 ID, 7/8 length, Turns ratio = 8 to 1
L_4	200 MHz RFC
R_1	100 Ω
R_2	1 kΩ

FIGURE 2. Small-signal power gain.

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NOTE 1: External interlead shield to isolate the collector from the emitter and base leads.

PROCEDURE:

1. Connect 31.8 MHz source to input connect on test jig.
2. Connect RF voltmeter to "CAL" connector on test jig.
3. Adjust 31.8 MHz signal level so that .5 volts is read on RF voltmeter.
4. Remove the RF voltmeter from "CAL" connector and connect to "OUT" connector.
5. Insert TUT into socket and adjust bias.
6. Read output on RF voltmeter and compute $r_b'C_c$ from the following relationship:

$$r_b'C_c\text{:(psec)} = 10 \times V_{eb} \text{ (millivolts)}$$

Derivation of $r_b'C_c$ readout:

h_{rb} set-up:

Set input (V_{cb}) at 1.0 Volt; then $h_{rb} = \frac{V_{eb}}{V_{cb}} = \frac{V_{eb}}{1}$ = meter reading.

$r_b'C_c = \frac{h_{rb}}{2\pi f}$; therefore, $h_{rb} = (2\pi f)(r_b'C_c)$, also $h_{rb} = \frac{V_{eb}}{V_{cb}}$.

Sample: at 31.8 MHz

Make $h_{rb} = .2$; $h_{rb} = \frac{V_{eb}}{V_{cb}} = \frac{.1}{.5} = .2$ then

$$r_b'C_c = \frac{h_{rb}}{2\pi f} = \frac{.2}{200 \times 10^6} = 1 \times 10^{-9} \times 1000 \text{ psec.}$$

(Other frequencies are treated in the same manner).

FIGURE 3. Collector-base time constant test circuit (an equivalent circuit may be used).

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For information purposes
Input levels (V_{cb}):

4.0 MHz = 400 mV	46.0 MHz = 346 mV
10.7 MHz = 471 mV	50.0 MHz = 318.4 mV
31.8 MHz = 500 mV	63.6 MHz = 250 mV
40.0 MHz = 400 mV	79.8 MHz = 200 mV

INTERPRETATION OF READINGS:

Meter range full scale	$r_b'C_c$ range (psec)		
	31.8, 40.0, 46.0, 50.0 63.6, 79.8 MHz	4.0 MHz	10.7 MHz
.001 volt	10 psec	100 psec	31.6
.003 volt	30 psec	300 psec	100.0
.01 volt	100 psec	1000 psec	316.0
.03 volt	300 psec	-	1000.0
.1 volt	1000 psec	-	-

FIGURE 3. Collector-base time constant test circuit (an equivalent circuit may be used). - continued

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Lead finish as specified (see 3.3.1).

b. Type designation and product assurance level.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Air Force - 19, 85, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/302B	2. DOCUMENT DATE (YYMMDD) 950119

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER, TYPES 2N2708, JAN

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6a. NAME OF SUBMITTER (Last, First, MI) (Optional)	6b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME DIRECTOR, US ARMY RESEARCH LABORATORY	b. TELEPHONE (Include Area Code) (1) Commercial: 908-544-3450 (2) AUTOVON: 995-3450
c. ADDRESS (Include Zip Code) POWER SOURCES DIRECTORATE ATTN: AMSRL-EP-RD FORT MONMOUTH, NJ 07703-5601	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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